CENG 252 Practice Midterm 2: Solutions

Lectures 1–10 • 100 points • 80 minutes

Problem	Final answers	Similar to (Fogler 5th ed.)
1	$\Delta H^\circ_{Rx} = -6000$ cal/mol, $\Delta C_P = 0$; $T = 300 + 120X$; at $X = 0.6$: $T = 372$ K, $k = 0.652$ min^{-1}, $K_C = 4.28$, $X_e = 0.81$, $V = 177$ dm^3; $X = 0.8$ impossible ($X_e(396\text{ K}) = 0.72$)	Ex. 11-2, 11-3(c), 11-4
2	$I_0/I = 1 + \dfrac{k_3}{k_2 C_M + k_4} C_Q$; $k' = 0.50$ L/mmol, $C_Q = 6.0$ mmol/L; $V_{\max} = 2.0$ mmol/(L·min), $K_M = 4.0$ mmol/L; $t = 27.2$ min	Ex. 9-1, 9-2, 9-3
3	$-r'_C = \dfrac{k(P_C - P_B P_P/K_P)}{1 + K_C P_C + K_B P_B}$; initial rates: linear, levelling off, constant; surface reaction limiting, $k = 0.50$ mol/(kg·s·atm), $K_C = 1.0$ atm^{-1}; $W = 30$ kg (20 kg if benzene is ignored)	§10.3, Ex. 10-2(b)
4	$r''_{\text{Dep}} = \dfrac{k' P_{\text{GeCl}_2} P_{\text{H}_2}}{\left(1 + K_A P_{\text{GeCl}_2} + \sqrt{K_H P_{\text{H}_2}}\right)^3}$; in GeCl_4: $\propto \dfrac{P_{\text{GeCl}_4} P_{\text{H}_2} P_{\text{Cl}_2}^2}{(\ldots)^3}$; $P^* = 0.25$ atm; orders $+1$ and -2	§10.5.2

How this exam maps to the instructor's announcement

- "One problem on energy balance (non-isothermal steady state)" → **Problem 1**. It is pitched at Ex. 11-2 (heat of reaction), Ex. 11-3(c) (adiabatic CSTR at the exit T) and the *strategy* of Ex. 11-4. As promised, it does not ask you to solve for the adiabatic equilibrium T and X.
- "Chapter 9 (9.1–9.3), Ex. 9-1, 9-2, 9-3" → **Problem 2**: a Stern–Volmer derivation, $V_{\max}$ and K_M from data, and a batch enzymatic reactor.
- "10.3 (benzene decomposition) is very important" → **Problem 3**, plus the hand-calculation part of Ex. 10-2 (fluidised CSTR). No X or P profiles are integrated, as promised.
- "10.5.2 (CVD) is very important" → **Problem 4**.

Timing: a well-prepared student needs about 18 + 13 + 20 + 14 = 65 minutes, which leaves about 15 minutes to check.

Problem 1: Energy balance, adiabatic CSTR (25 pts)

(a) Heat of reaction

$$\Delta H^\circ_{Rx} = H^\circ_B - H^\circ_A = -41{,}000 - (-35{,}000) = -6000 \text{ cal/mol A} \quad \text{(exothermic)},$$

$$\Delta C_P = C_{P,B} - C_{P,A} = 40 - 40 = 0.$$

Answer

$\Delta H^\circ_{Rx} = -6000$ cal/mol A. Since $\Delta C_P = 0$, $\Delta H_{Rx}(T) = \Delta H^\circ_{Rx}(T_R) + \Delta C_P(T - T_R) = -6000$ cal/mol at every temperature.

(b) Adiabatic T–X relation

Step 1. Adiabatic ($\dot{Q} = 0$), no shaft work ($\dot{W}_s = 0$), all species fed at T_0, and $\Delta C_P = 0$:

$$-F_{A0}\sum \Theta_i C_{P,i}(T - T_0) - \Delta H^\circ_{Rx} F_{A0} X = 0 \quad \Rightarrow \quad T = T_0 + \frac{(-\Delta H^\circ_{Rx})X}{\sum \Theta_i C_{P,i}}.$$

Step 2. $\Theta_B = 0$ and $\Theta_I = 0.20/0.80 = 0.25$, so $\sum \Theta_i C_{P,i} = C_{P,A} + \Theta_I C_{P,I} = 40 + 0.25(40) = 50$ cal/(mol A·K).

Answer

$$T = 300 + \frac{6000}{50}X = 300 + 120X \text{ (K)}$$

Common mistake

Leaving out the inert ($\sum \Theta_i C_{P,i} = 40$ gives $T = 300 + 150X$, which is too hot). Θ_I is per mole of A fed: 0.20/0.80, not 0.20.

(c) CSTR volume for $X = 0.60$

A CSTR operates entirely at its exit conditions, so evaluate everything at $X = 0.60$.

Step 1. Exit temperature: $T = 300 + 120(0.60) = 372$ K. $\quad \dfrac{1}{300} - \dfrac{1}{372} = 6.452 \times 10^{-4}$ K^{-1}.

Step 2. Rate constant: $k = 0.005 \exp\left[\dfrac{15{,}000}{1.987}(6.452 \times 10^{-4})\right] = 0.005\, e^{4.870} = 0.652$ min^{-1}.

Step 3. Equilibrium constant (exothermic, so K_C falls): $K_C = 30 \exp\left[\dfrac{-6000}{1.987}(6.452 \times 10^{-4})\right] = 30\, e^{-1.948} = 4.28$.

Step 4. Feasibility check: $X_e = \dfrac{K_C}{1 + K_C} = \dfrac{4.28}{5.28} = 0.81 > 0.60$. ✓

Step 5. Liquid-phase stoichiometry: $C_A = C_{A0}(1 - X) = 0.80$ and $C_B = C_{A0}X = 1.20$ mol/dm^3:

$$-r_A = k\left(C_A - \frac{C_B}{K_C}\right) = 0.652\left(0.80 - \frac{1.20}{4.28}\right) = 0.652(0.520) = 0.339 \text{ mol/(dm}^3 \cdot \text{min)}.$$

Step 6. $F_{A0} = 0.80(125) = 100$ mol/min:

$$V = \frac{F_{A0}X}{-r_A} = \frac{(100)(0.60)}{0.339}.$$

Answer

$V \approx$ **177 dm^3** (with $v_0 = F_{A0}/C_{A0} = 50$ dm^3/min, $\tau = 3.5$ min), operating at 372 K. For comparison, an *isothermal* CSTR at 300 K would need about 15,800 dm^3. The heat released by the reaction speeds it up roughly 90-fold.

Common mistake

(1) Using k and K_C at the *feed* temperature. (2) Using the X_e of the adiabatic equilibrium point instead of X_e at the reactor temperature. (3) Writing $-r_A = kC_A$ and dropping the reverse term.

(d) The adiabatic limit

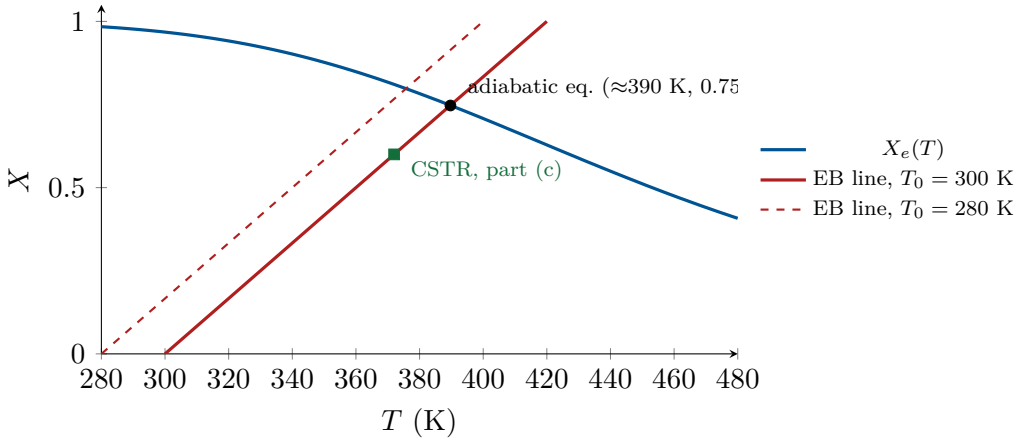
(i) At $X = 0.80$ the energy balance gives $T = 300 + 120(0.80) = 396$ K. There,

$$K_C = 30\exp\left[\frac{-6000}{1.987}\left(\frac{1}{300}-\frac{1}{396}\right)\right] = 30\,e^{-2.440} = 2.61, \qquad X_e = \frac{2.61}{3.61} = 0.72.$$

Answer

(i) To reach $X = 0.80$, the adiabatic feed must heat up to 396 K. At that temperature equilibrium allows only $X_e = 0.72 < 0.80$, so **no adiabatic reactor of any size can reach 0.80**. (For reference, the adiabatic equilibrium point is at about 390 K and $X \approx 0.75$; you were not asked to find it.)
(ii) The equilibrium curve $X_e(T)$ falls with T (exothermic). The energy-balance line rises from $(T_0, 0)$ with slope $\sum \Theta_i C_{P,i}/(-\Delta H^\circ_{Rx})$. They intersect at the adiabatic equilibrium point. Ways to go higher:

- a **lower feed temperature** T_0 (the line shifts left; $T_0 = 280$ K gives about 0.80);
- **more inert** (larger $\sum \Theta_i C_{P,i}$, so the line is steeper and T rises less);
- **reactors in series with interstage cooling**, or removing heat (a non-adiabatic reactor).



Grading guide

(a) 2 for ΔH°_{Rx} with its sign and basis, 1 for $\Delta C_P = 0$ and the consequence. (b) 2 for reducing the energy balance to the adiabatic form, 2 for Θ_I and $\sum \Theta_i C_{P,i}$, 2 for $T = 300 + 120X$. (c) 1 for T, 2 for k, 2 for K_C, 1 for the X_e check, 2 for the rate with stoichiometry, 2 for V. (d) 3 for (i) with the $X_e(396)$ calculation, 3 for (ii): sketch, intersection and two valid fixes.

Problem 2: Reaction pathways and enzymes (20 pts)

(a) Stern–Volmer equation

Step 1. Write the rate of every step for D*: formation = I_a; loss by (2) = $k_2 C_M C_{D^*}$; by (3) = $k_3 C_Q C_{D^*}$; by (4) = $k_4 C_{D^*}$.

Step 2. PSSH on D*:

$$0 = I_a - k_2 C_M C_{D^*} - k_3 C_Q C_{D^*} - k_4 C_{D^*} \quad \Rightarrow \quad C_{D^*} = \frac{I_a}{k_2 C_M + k_3 C_Q + k_4}.$$

Step 3. Intensity with and without quencher:

$$I = k_4 C_{D^*} = \frac{k_4 I_a}{k_2 C_M + k_3 C_Q + k_4}, \qquad I_0 = I(C_Q = 0) = \frac{k_4 I_a}{k_2 C_M + k_4}.$$

Step 4. Take the ratio:

Answer

$$\frac{I_0}{I} = \frac{k_2 C_M + k_3 C_Q + k_4}{k_2 C_M + k_4} = 1 + k' C_Q, \qquad k' = \frac{k_3}{k_2 C_M + k_4}$$

C_M (the solvent) is constant, so the Stern–Volmer plot of I_0/I against C_Q is a straight line with intercept 1. I_a (and k_4) cancel because the same lamp excites the dye in both measurements: the ratio isolates the competition for D*.

(b) Numbers

Answer

$k' = \dfrac{2.5 - 1}{3.0} = \mathbf{0.50\ L/mmol}$. For $I = I_0/4$: $I_0/I = 4 = 1 + 0.50\, C_Q \Rightarrow C_Q = \mathbf{6.0\ mmol/L}$.

(c) Lineweaver–Burk

$1/C_S$ (L/mmol)	1.000	0.500	0.250	0.125
$1/(-r_S)$ (L·min/mmol)	2.50	1.50	1.00	0.75

Slope $= \dfrac{2.50 - 0.75}{1.000 - 0.125} = 2.0$ min; intercept $= 2.50 - 2.0(1.000) = 0.50$ L·min/mmol.

Answer

$V_{\max} = 1/0.50 = \mathbf{2.0\ mmol/(L{\cdot}min)}$; $K_M = \text{slope} \times V_{\max} = 2.0 \times 2.0 = \mathbf{4.0\ mmol/L}$.

(d) Batch reactor at half the enzyme loading

Step 1. $V_{\max} = k_{\text{cat}} C_{E,t}$ is proportional to the enzyme loading, so at 0.50 g/L, $V_{\max} = 1.0$ mmol/(L·min). $K_M = (k_2 + k_3)/k_1$ is a ratio of rate constants and does **not** depend on the amount of enzyme: $K_M = 4.0$ mmol/L.

Step 2. Mole balance: $\dfrac{dC_S}{dt} = -\dfrac{V_{\max} C_S}{K_M + C_S} \Rightarrow -\displaystyle\int_{C_{S0}}^{C_S} \left(\frac{K_M}{C_S} + 1\right) dC_S = V_{\max} t$, so

$$t = \frac{K_M}{V_{\max}} \ln \frac{1}{1-X} + \frac{C_{S0} X}{V_{\max}}.$$

Step 3. $t = \dfrac{4.0}{1.0} \ln 10 + \dfrac{(20)(0.90)}{1.0} = 9.2 + 18.0.$

Answer

$t = \mathbf{27.2\ min}$. Halving the enzyme loading halves $V_{\max}$ and doubles the time; K_M is unchanged.

Common mistake

Scaling K_M with the enzyme loading, or using the CSTR formula $t = C_{S0}X/(-r_S)$ for a batch reactor.

Grading guide

(a) 2 for the step rates, 2 for the PSSH and C_{D^*}, 2 for the ratio, 1 for why I_a cancels. (b) 1.5 + 1.5. (c) 2 for the reciprocals and fit, 2 for $V_{\max}$ and K_M with units. (d) 2 for the scaling argument, 2 for the integrated batch equation, 2 for 27.2 min.

Problem 3: Cumene decomposition (30 pts)

(a) Surface reaction limiting

Step 1. RLS rate law. The forward step is single-site; the reverse is P from the gas attacking B·S:

$$-r'_C = r_S = k_S \left(C_{\text{CS}} - \frac{P_P C_{\text{BS}}}{K_S}\right).$$

Step 2. The fast steps are at equilibrium:

$$\frac{r_{AD}}{k_A} = P_C C_v - \frac{C_{\text{CS}}}{K_C} \approx 0 \Rightarrow C_{\text{CS}} = K_C P_C C_v; \qquad \frac{r_D}{k_D} = C_{\text{BS}} - K_B P_B C_v \approx 0 \Rightarrow C_{\text{BS}} = K_B P_B C_v.$$

Step 3. Substitute: $r_S = k_S C_v \left(K_C P_C - \dfrac{K_B P_B P_P}{K_S}\right) = k_S K_C C_v \left(P_C - \dfrac{P_B P_P}{K_P}\right)$, using $K_P = \dfrac{K_C K_S}{K_B}$.

Step 4. Site balance (P is not adsorbed): $C_t = C_v + C_{\text{CS}} + C_{\text{BS}} = C_v(1 + K_C P_C + K_B P_B)$.

Answer

$$-r'_C = \frac{k\,(P_C - P_B P_P / K_P)}{1 + K_C P_C + K_B P_B}, \qquad k = k_S K_C C_t$$

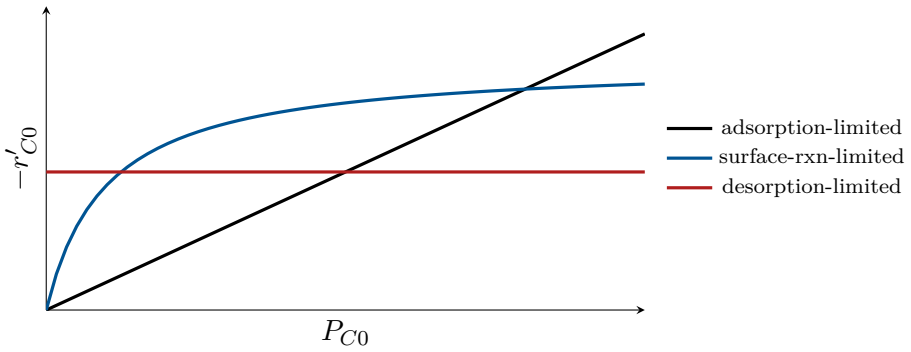
There is no P_P term in the denominator, because propylene does not adsorb.

(b) Initial rates

Pure cumene feed: $P_B = P_P = 0$ at the inlet.

Answer

- Adsorption-limited: $-r'_{C0} = k_A C_t P_{C0}$, **linear**, with no levelling off.
- Surface-reaction-limited: $-r'_{C0} = \dfrac{kP_{C0}}{1 + K_C P_{C0}}$, linear at low P_{C0}, then **levels off** at k/K_C.
- Desorption-limited: with $P_P = 0$ the denominator is just $K_S K_C P_{C0}$, so $-r'_{C0} = k_D C_t$, **independent of** P_{C0}. With no propylene, B·S cannot react back, the surface fills with B, and the rate is set by how fast B leaves.



Common mistake

Writing the desorption-limited initial rate as proportional to P_{C0}. You must set $P_P = 0$ *before* simplifying: the P_P terms vanish from the denominator, and P_{C0} cancels.

(c) Identifying the RLS and its parameters

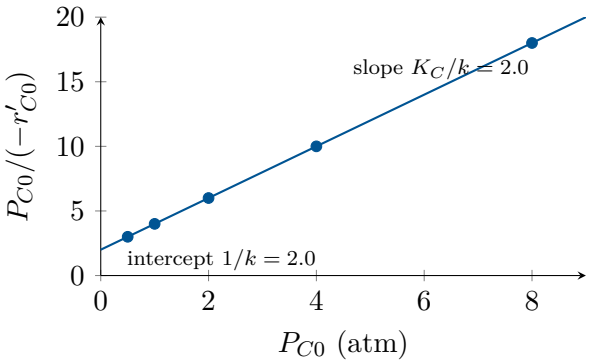
The rate rises and levels off, so the **surface reaction is rate-limiting**, as Fogler found for the real cumene data. Linearise: $\dfrac{P_{C0}}{-r'_{C0}} = \dfrac{1}{k} + \dfrac{K_C}{k} P_{C0}$.

P_{C0} (atm)	0.5	1.0	2.0	4.0	8.0
$P_{C0}/(-r'_{C0})$ (atm·kg·s/mol)	3.0	4.0	6.0	10.0	18.0

Slope $= \dfrac{18.0 - 3.0}{8.0 - 0.5} = 2.0$ kg·s/mol; intercept $= 3.0 - 2.0(0.5) = 2.0$ atm·kg·s/mol.

Answer

$k = 1/2.0 =$ **0.50 mol/(kg cat·s·atm)**; $K_C =$ slope $\times\, k =$ **1.0 atm^{-1}**.



(d) Fluidised CSTR

Step 1. Stoichiometry: pure C, C $\longrightarrow$ B + P: $\delta = 1$, $\varepsilon = 1$. Isothermal, no ΔP:

$$P_C = P_0\frac{1-X}{1+X} = 3\cdot\frac{0.5}{1.5} = 1.0 \text{ atm}, \qquad P_B = P_P = P_0\frac{X}{1+X} = 1.0 \text{ atm}.$$

Step 2. Exit rate (irreversible form): $-r'_C = \dfrac{kP_C}{1+K_CP_C+K_BP_B} = \dfrac{0.5(1.0)}{1+1.0+1.0} = 0.167$ mol/(kg·s).

Step 3. $W = \dfrac{F_{C0}X}{-r'_C} = \dfrac{(10)(0.5)}{0.167} = 30$ kg.

Answer

$W =$ **30 kg**. Ignoring benzene: $-r'_C = 0.5/2 = 0.25$, so $W = 20$ kg, an underestimate by a **factor of 1.5**. Benzene, the product, occupies sites that cumene needs (product inhibition). The CSTR runs entirely at the exit composition, where P_B is high, so the effect is large.

Common mistake

Using $P_C = P_0(1-X)$ (that is, $\varepsilon = 0$). One mole of gas becomes two, so the mole fraction of cumene is diluted by the factor $1+X$.

Grading guide

(a) 2 for the RLS rate law with the reverse term, 3 for the equilibrium expressions, 2 for the site balance, 1 for the final form. (b) 1 for each initial rate, 3 for the sketch. (c) 2 for the identification with a reason, 3 for the linearisation and table, 3 for k and K_C with units. (d) 3 for ε and the partial pressures, 2 for the rate, 2 for W, 1 for the factor 1.5 with the reason.

Problem 4: CVD of germanium (25 pts)

(a) Deposition rate

Step 1. RLS: $r''_{\text{Dep}} = k_S f_{\text{GeCl}_2} f_{\text{H}}^2$.

Step 2. GeCl_2 adsorption at equilibrium:

$$r_{AD} = k_A\left(f_v P_{\text{GeCl}_2} - \frac{f_{\text{GeCl}_2}}{K_A}\right) \approx 0 \;\Rightarrow\; f_{\text{GeCl}_2} = K_A P_{\text{GeCl}_2} f_v.$$

Step 3. Dissociative H_2 adsorption at equilibrium (two sites in, two H atoms out):

$$r_{AD,H_2} = k_H \left(P_{H_2} f_v^2 - \frac{f_H^2}{K_H} \right) \approx 0 \;\Rightarrow\; f_H = \sqrt{K_H P_{H_2}}\, f_v.$$

Step 4. Substitute: $r''_{Dep} = k_S K_A K_H P_{GeCl_2} P_{H_2} f_v^3$.

Step 5. Area balance: $1 = f_v + f_{GeCl_2} + f_H = f_v \left(1 + K_A P_{GeCl_2} + \sqrt{K_H P_{H_2}}\right)$.

Answer

$$r''_{Dep} = \frac{k' P_{GeCl_2} P_{H_2}}{\left(1 + K_A P_{GeCl_2} + \sqrt{K_H P_{H_2}}\right)^3}, \qquad k' = k_S K_A K_H$$

(Fogler Eq. 10-90.)

(b) In terms of $GeCl_4$

The gas-phase equilibrium gives $K_P = \dfrac{P_{GeCl_2} P_{Cl_2}}{P_{GeCl_4}}$, so $P_{GeCl_2} = \dfrac{K_P P_{GeCl_4}}{P_{Cl_2}}$. Substitute, then multiply the top and bottom by $P_{Cl_2}^3$:

Answer

$$r''_{Dep} = \frac{k' K_P\, P_{GeCl_4} P_{H_2} P_{Cl_2}^2}{\left[P_{Cl_2}\left(1 + \sqrt{K_H P_{H_2}}\right) + K_A K_P P_{GeCl_4}\right]^3}$$

If H_2 is weakly adsorbed this becomes $\dfrac{k P_{GeCl_4} P_{H_2} P_{Cl_2}^2}{\left(P_{Cl_2} + K_A K_P P_{GeCl_4}\right)^3}$ (Fogler Eq. 10-91). **Effect of Cl_2:** by Le Chatelier, Cl_2 pushes the gas-phase equilibrium back towards $GeCl_4$, which lowers P_{GeCl_2}, the species that actually adsorbs and deposits. At low $GeCl_2$ coverage the rate goes as $1/P_{Cl_2}$.

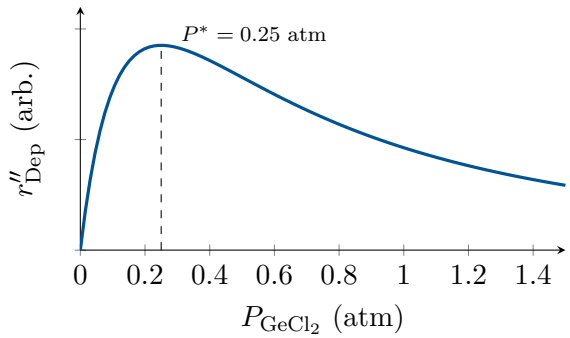
(c) Optimum $GeCl_2$ pressure

Let $a = 1 + \sqrt{K_H P_{H_2}} = 2.0$ and $P = P_{GeCl_2}$, so $r'' \propto \dfrac{P}{(a + K_A P)^3}$.

$$\frac{d}{dP}\left[\frac{P}{(a + K_A P)^3}\right] = \frac{(a + K_A P) - 3K_A P}{(a + K_A P)^4} = 0 \;\Rightarrow\; P^* = \frac{a}{2K_A} = \frac{2.0}{2(4.0)}.$$

Answer

$P^*_{GeCl_2} = \mathbf{0.25\ atm}$. Apparent order in $GeCl_2$: $+1$ at low pressure ($K_A P \ll a$), and -2 at high pressure ($r'' \propto P/P^3$). Beyond P^*, $GeCl_2$ crowds out the H atoms it needs as reaction partners.



(d) Concepts

Answer

(i) The RLS needs **three** sites at once (one $GeCl_2$·S and two H·S), so the rate contains f_v^3, and the area balance turns that into a cubed denominator.

(ii) H_2 adsorbs **dissociatively** into two H atoms, so $f_H^2 \propto P_{H_2} f_v^2$, giving $f_H \propto \sqrt{P_{H_2}}$.

(iii) Three occupied sites react, and two vacant sites are released, but the newly deposited Ge atom is **itself a new surface site**: three sites before, three after. The area balance stays valid as the film grows.

Common mistake

(1) $f_H = K_H P_{H_2} f_v$ (no square root). (2) A squared denominator (counting two sites instead of three). (3) Writing the rate in terms of $GeCl_4$ directly: $GeCl_4$ never adsorbs, only $GeCl_2$ does.

Grading guide

(a) 2 for the RLS rate, 2 for f_{GeCl_2}, 3 for f_H with the square root, 2 for the area balance, 1 for the final form. (b) 3 for the substitution, 2 for the Cl_2 explanation. (c) 3 for the derivative and P^*, 1 for the number, 2 for both orders. (d) 1 + 1 + 2.